

CE 310

Simple Linear Regression

Week 9 · Session 1 · From Correlation to Prediction

University of Arizona · Dr. Wooyoung Jung · Fall 2026

The Number Goes in the Contract

Your firm is writing an **energy performance guarantee** for the Tucson medium office. The client asks one question before signing:

*"Our summers keep running hotter. If the cooling season averages **3°F above** the design year, how much more electricity will this building use?"*

From Week 7 you have one number, computed from 8,760 hours of this building's own data:

$$r(\text{Outdoor_Temp_F}, \text{Cooling_kWh}) = 0.78$$

⚠ The guarantee has to state a figure in **kWh**. Can you produce one from $r = 0.78$?

✓ No — and the Reason Is Units

What $r = 0.78$ gives you

- Direction: positive ✓
- Strength: strong ✓
- Dimensionless — travels across any units ✓

What it cannot give you

- kWh per 1°F rise
- Demand at the 108°F design temperature
- Where the chiller shuts off

What this week produces instead:

$$\beta_1 = 1.1585 \text{ kWh}/^\circ\text{F}$$

"For every 1°F rise in outdoor temperature, predicted cooling demand rises by 1.16 kWh/hr."

A 3°F warmer season is **3.5 kWh/hr across every cooling hour** — multiply by the season's hours and the utility rate, and you have a figure you can sign.

💡 r answers *does it move together*. β_1 answers *by how much* — in units you can quote.

Today's Roadmap

By the end of this lecture you will be able to:

1. Explain why correlation alone is insufficient for engineering decisions
2. Write and interpret the regression equation $\hat{y} = \beta_0 + \beta_1 x$
3. Interpret slope and intercept in physical units
4. Read R^2 as the fraction of variance explained
5. Use a residual plot to identify where a linear model breaks down
6. Connect linear regression to supervised machine learning

Dataset: DOE Reference Medium Office, Tucson — 8,760 hourly observations (the Week 7 file)

Python Readiness — What You Need This Week

Introduced in this week's on-ramp:

- `pandas` — read CSV, column access, filtering
- `matplotlib.pyplot` — scatter plots, line plots, labels
- `numpy` — array arithmetic, `mean()`, `std()`

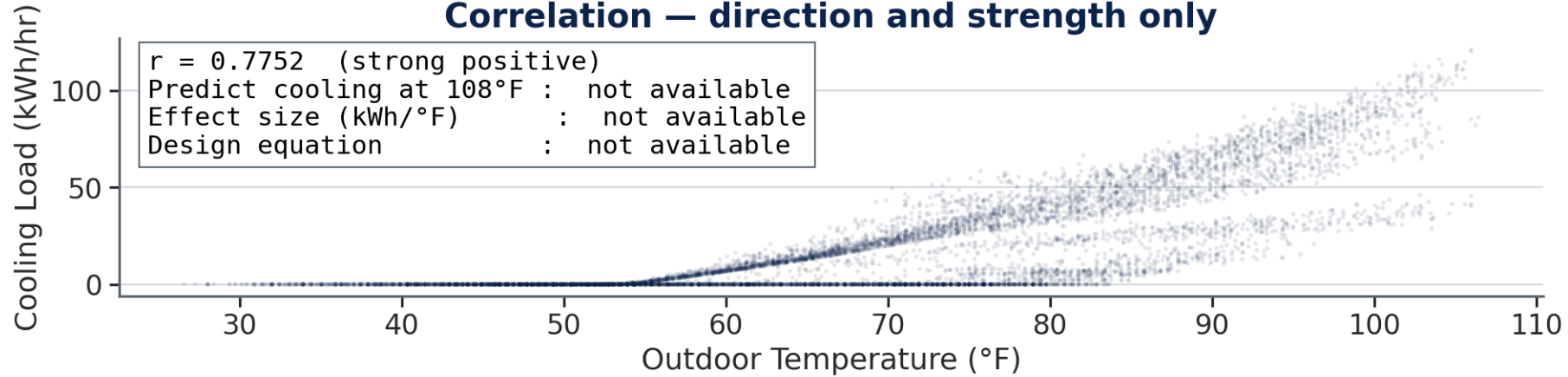
New this week:

- `scipy.stats.linregress` — fits OLS line; returns slope, intercept, r, p-value, SE
- `statsmodels.formula.api` — prediction intervals (tutorial only)

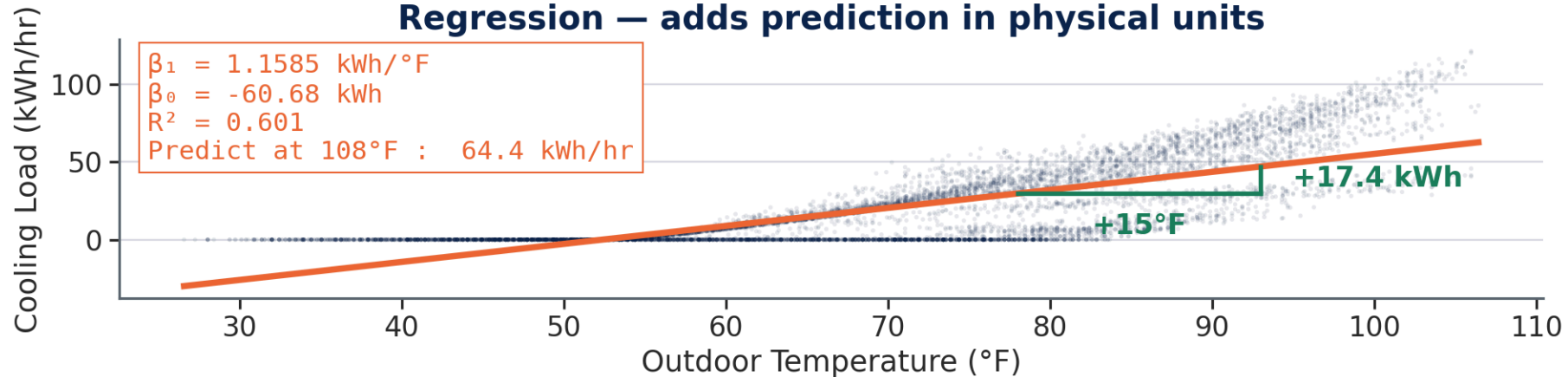
No prior programming assumed. Work the Quickstart notebook before the tutorial — it is where the three tools on the left are introduced. Never coded at all? Start with the Foundations notebook ahead of it.

From Correlation to Prediction

Correlation — direction and strength only



Regression — adds prediction in physical units



The Regression Equation

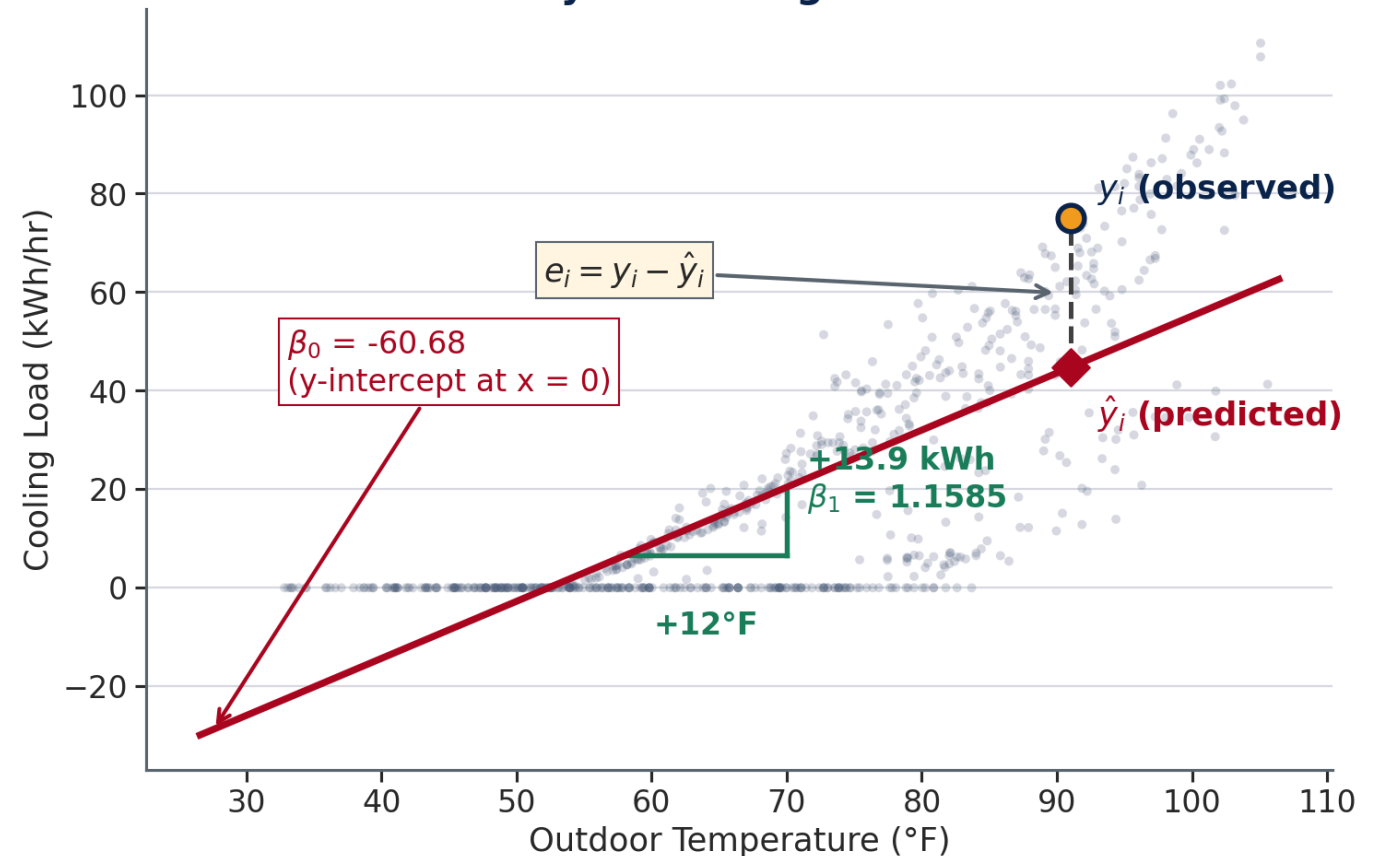
$$\hat{y} = \beta_0 + \beta_1 \cdot x$$

- $\hat{y}$ — predicted value (not the observed value)
- β_1 — slope: change in $\hat{y}$ per 1-unit rise in x
- β_0 — intercept: predicted $\hat{y}$ when $x = 0$
- x — predictor (input) variable

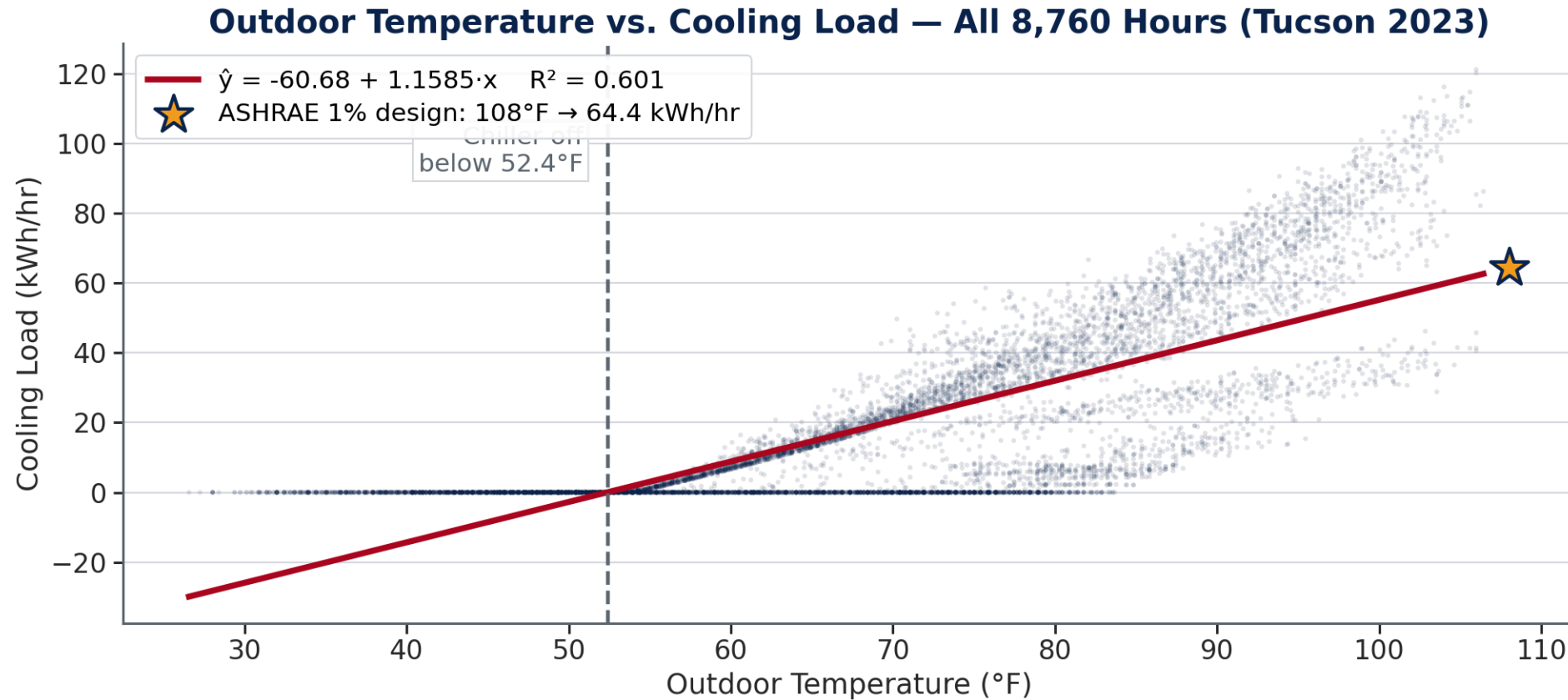
Tucson cooling model:

$$\widehat{\text{Cooling}} = -60.68 + 1.1585 \times T_{\text{out}}$$

Anatomy of the Regression Line



The Fitted Model — 8,760 Hours



$\beta_1 = 1.1585 \text{ kWh/°F} \cdot R^2 = 0.601 \cdot n = 8,760 \text{ hrs} \cdot \text{DOE Reference Medium Office, Tucson AZ 2023}$

Interpreting the Slope

$$\beta_1 = 1.1585 \text{ kWh/}^\circ\text{F}$$

"For every 1°F rise in outdoor temperature, predicted cooling demand increases by **1.16 kWh/hr.**"

Engineering applications of the slope:

Scenario	Calculation	Result
10°F hotter design day	1.1585×10	+11.6 kWh/hr more cooling
Summer vs. mild day (+20°F)	1.1585×20	+23.2 kWh/hr more cooling
Climate zone shift (+5°F mean)	1.1585×5	+5.8 kWh/hr annual load increase

The slope is the number a mechanical engineer needs to size cooling equipment for different climate conditions.

r vs. β_1 — Two Different Questions

	Pearson correlation r	Regression slope β_1
Measures	Strength and direction	Rate of change
Units	Unitless (-1 to $+1$)	kWh / °F (physical units)
Question answered	Is there a relationship?	How much does y change per °F?
Tucson example	$r = 0.78$	$\beta_1 = 1.16$ kWh / °F

A strong r does not imply a steep β_1 — and vice versa.

r tells you the relationship is real and strong.

β_1 tells you what to do with it — write a design equation, size equipment, predict demand.

Interpreting the Intercept

$$\beta_0 = -60.68 \text{ kWh}$$

The intercept is "predicted cooling when outdoor temperature = 0°F."

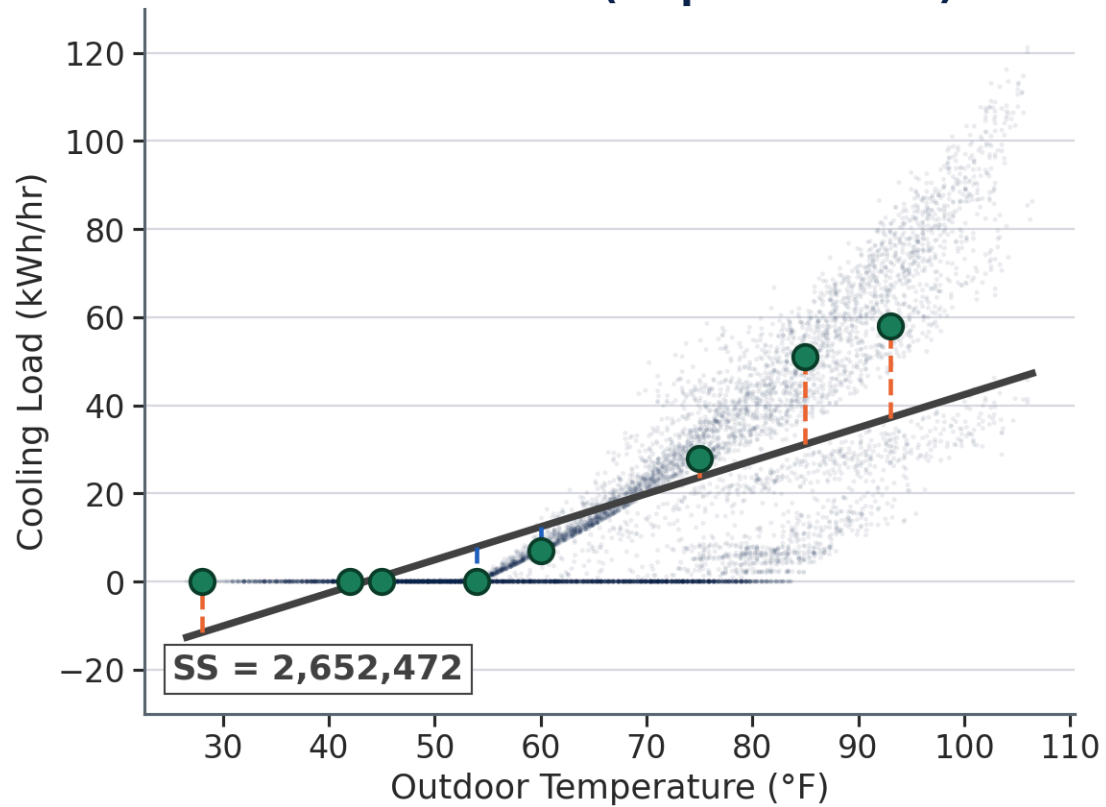
- Tucson's minimum observed temperature: **26.6°F** — never reaches 0°F
- The intercept is mathematically valid but **physically meaningless** for this dataset
- A negative intercept simply reflects that the chiller is off in winter

⚠ Never interpret an intercept at $x = 0$ if 0 is outside your observed data range. Extrapolation is unreliable and often physically absurd.

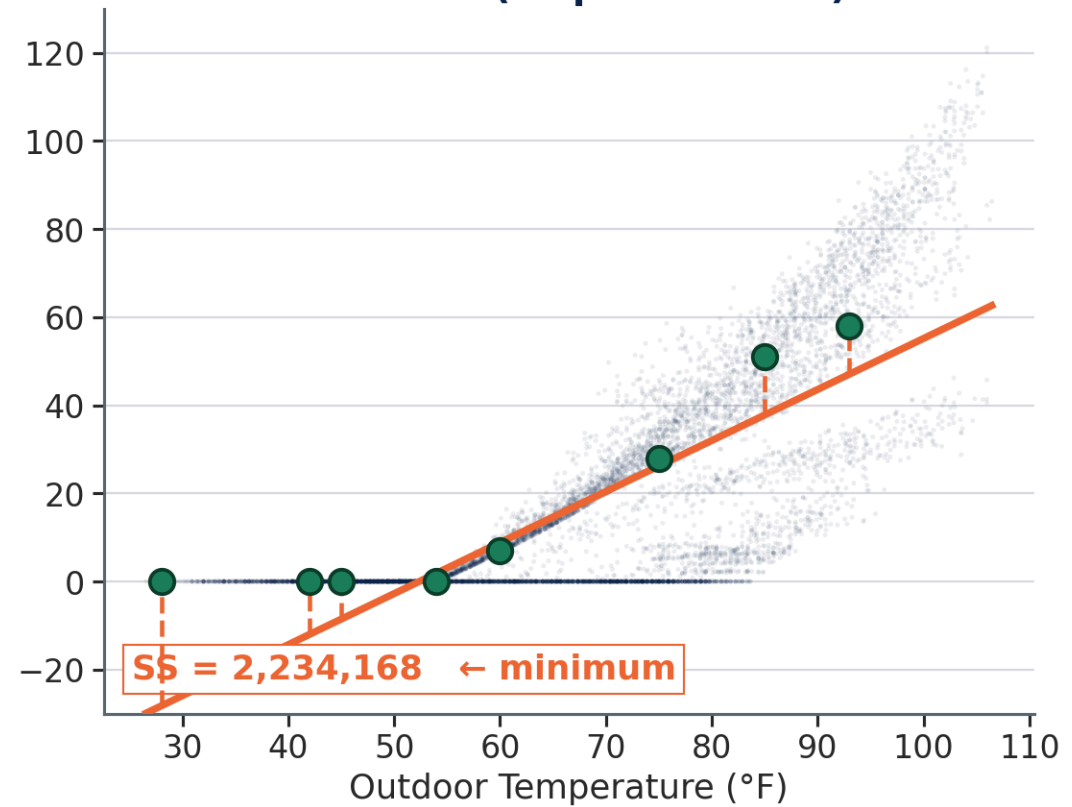
OLS Minimizes Total Squared Error

OLS minimizes $\sum e_i^2$ — the sum of squared vertical distances ($n = 8,760$)

Candidate line (slope = 0.7500)



OLS line (slope = 1.1585)



How the Line Is Found — Least Squares

Choose β_0 and β_1 to **minimize total squared error** — where $e_i = y_i - \hat{y}_i$:

$$\min_{\beta_0, \beta_1} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2$$

Why squaring? A 10 kWh miss counts 100× more than a 1 kWh miss; squaring also yields a unique closed-form solution (minimizing $|e_i|$ does not).

Week 7 connection — closed-form solution:

$$\hat{\beta}_1 = r \cdot \frac{\sigma_Y}{\sigma_X} = \frac{\text{cov}(X, Y)}{\text{var}(X)}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

The line always passes through $(\bar{x}, \bar{y})$. Verify: $-60.68 + 1.1585 \times 68.7 = 19.0$ kWh/hr ✓

Quick Gut Check — r to R^2

From Week 7: $r = 0.78$ between outdoor temperature and cooling demand.

What fraction of cooling variance does outdoor temperature explain?

A) $\approx 20\%$ **B)** $\approx 40\%$ **C)** $\approx 60\%$ **D)** $\approx 80\%$

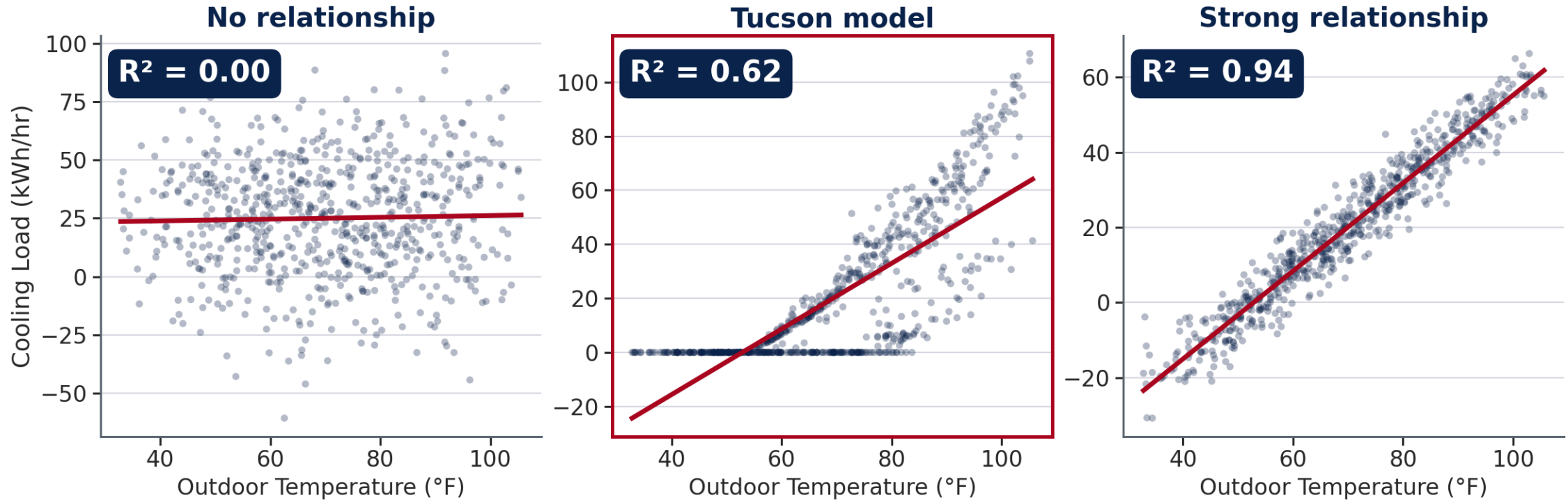
Work it out — you have the number you need from Week 7.

✓ $R^2 = r^2 = 0.78^2 \approx 0.60$

Outdoor temperature explains **60%** of cooling variance.

- The remaining **40%** — occupancy patterns, solar gain, equipment loads — is not captured by temperature alone
- $R^2 = r^2$ holds **only for simple linear regression** (one predictor); in Week 10 with two predictors it no longer applies
- Adding business-hours as a second predictor next week pushes R^2 to **0.91** during occupied hours

R^2 — Fraction of Variance Explained

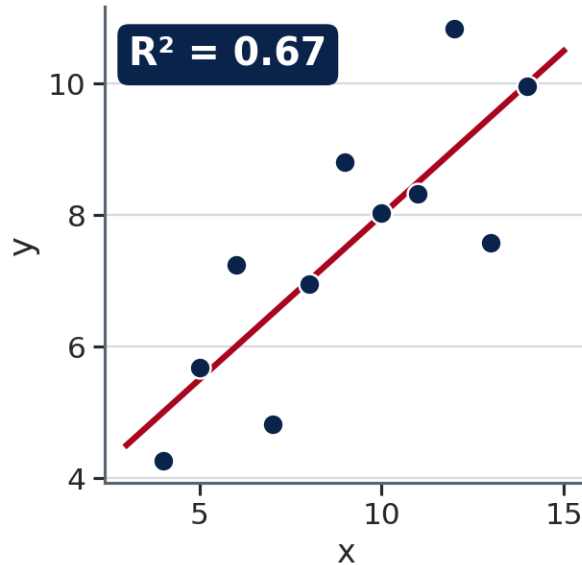


Tucson: $R^2 = 0.601$ — outdoor temperature explains 60% of cooling variance; the rest is occupancy, solar, and equipment.

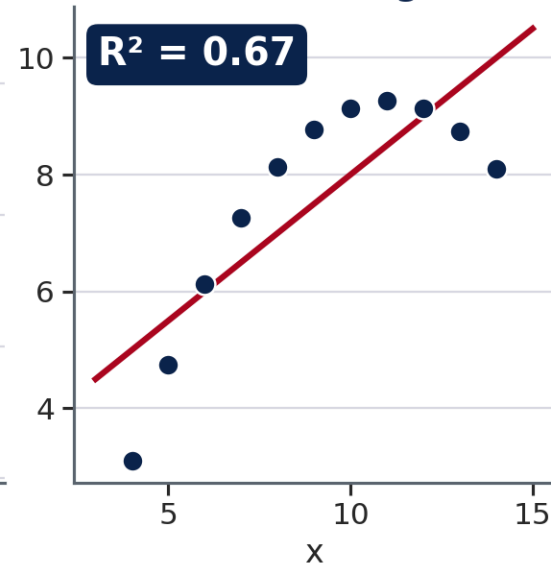
Anscombe's Quartet — Same Statistics, Different Relationships

Anscombe's Quartet — nearly identical statistics, completely different relationships

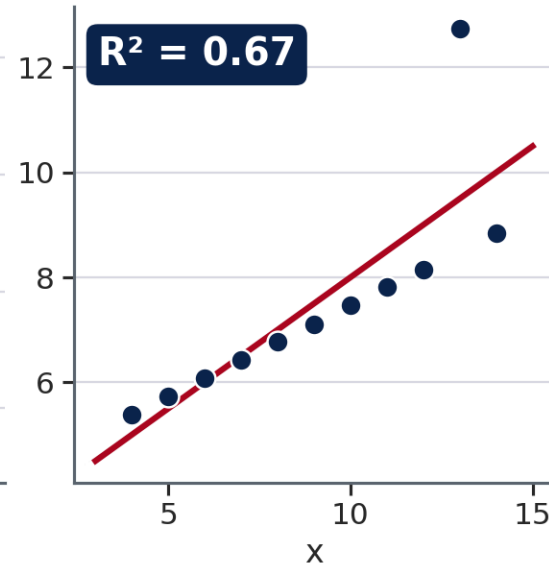
Dataset I:
Ideal — linear



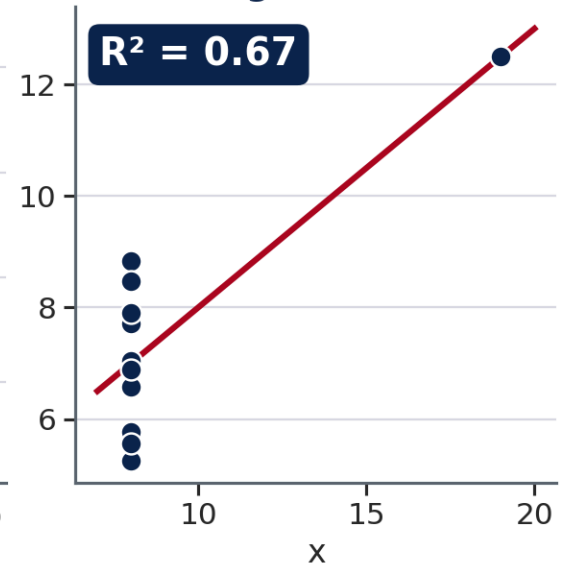
Dataset II:
Curved — wrong form



Dataset III:
Outlier — distorts line



Dataset IV:
Leverage — no trend



What Anscombe's Quartet Tells Us

All four datasets share nearly identical statistics:

slope ≈ 0.50 , intercept ≈ 3.0 , $R^2 \approx 0.67$

Summary statistics alone never replace a scatter plot. Always visualize your data before fitting any model.

R^2 cannot tell you that your model is wrong. Only the scatter plot and residual plot can.

- **Dataset I** — the only one where OLS is appropriate
- **Dataset II** — curved relationship; OLS underpredicts in the middle, overpredicts at the extremes
- **Dataset III** — one outlier inflates the slope; remove it and $R^2 > 0.99$
- **Dataset IV** — no real relationship; a single leverage point creates the illusion of a trend

Regression Assumptions — LINE

For OLS estimates to be valid and confidence intervals to be trustworthy, four conditions must hold:

- **L — Linearity**: the true relationship between x and y is linear
- **I — Independence**: each observation is independent (no serial correlation in residuals)
- **N — Normality**: residuals follow an approximately normal distribution
- **E — Equal variance**: residual spread is constant across all values of x (homoscedasticity)

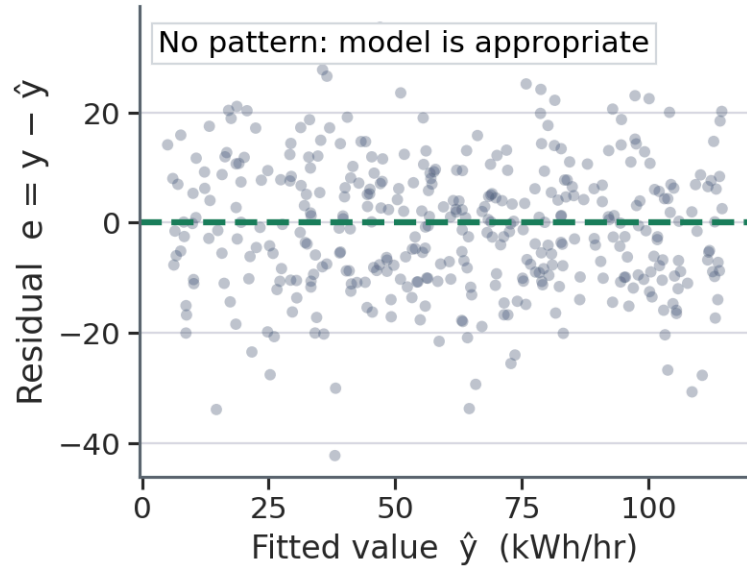
The residual patterns figure (next slide) shows what **L** and **E** violations look like in practice.

The Tucson hourly data has mild **I** violations (consecutive hours, same building) — a limitation accepted at this course level.

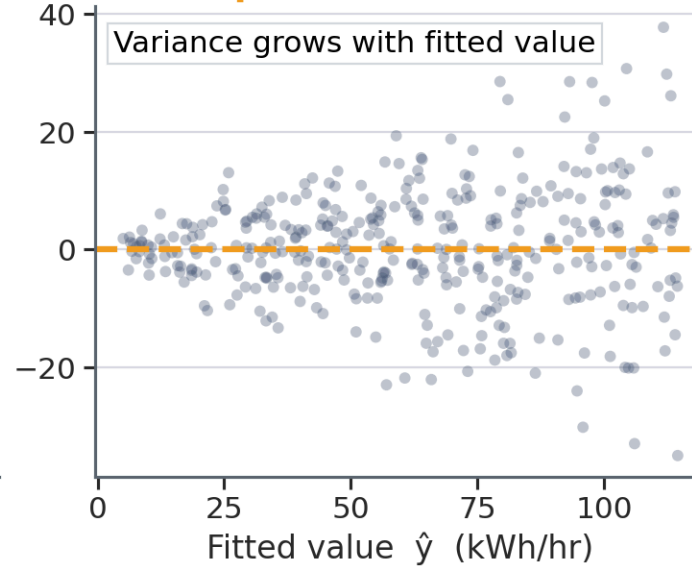
Residual Patterns — What to Look For

Residual plot patterns — what each shape tells you about the model

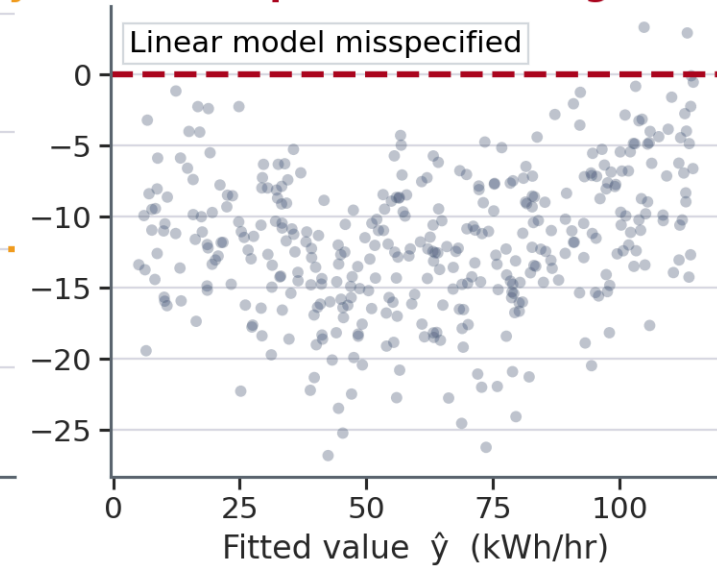
Good — random scatter



Fan shape — heteroscedasticity



Curved pattern — wrong form



Residuals — Checking the Model

Residual: $e_i = y_i - \hat{y}_i$

- **Positive residual:** actual > predicted (model underestimates)
- **Negative residual:** actual < predicted (model overestimates)

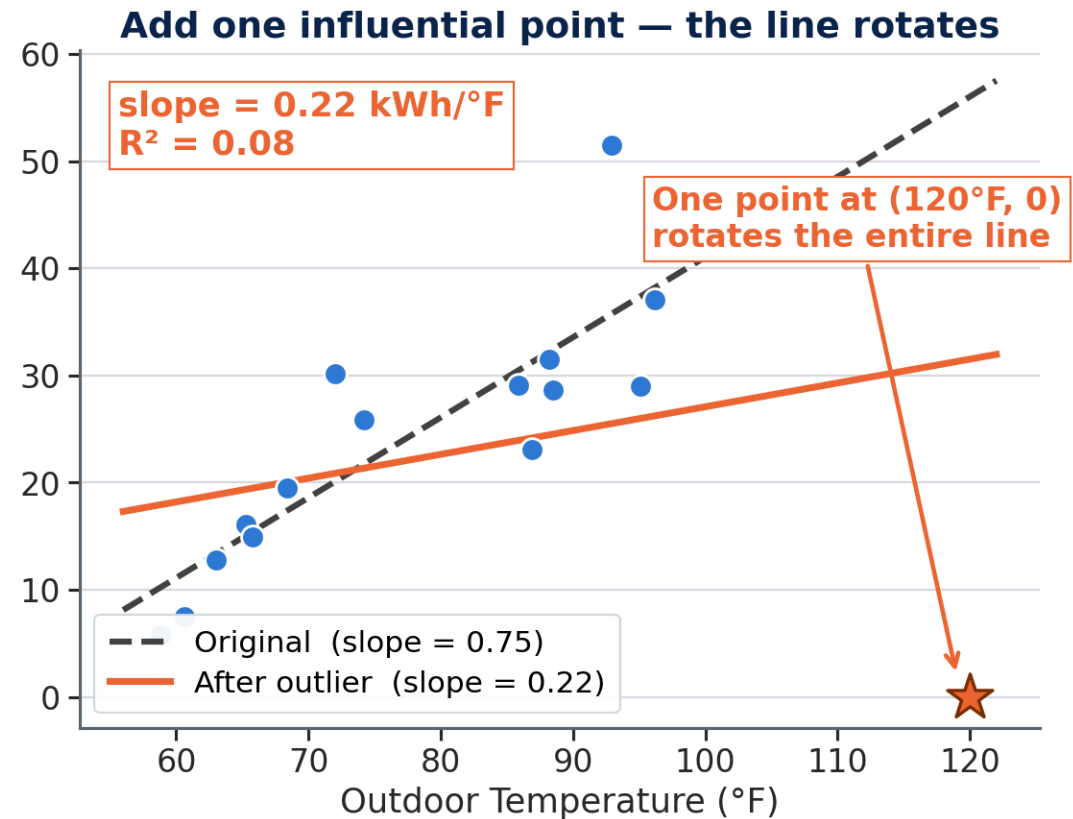
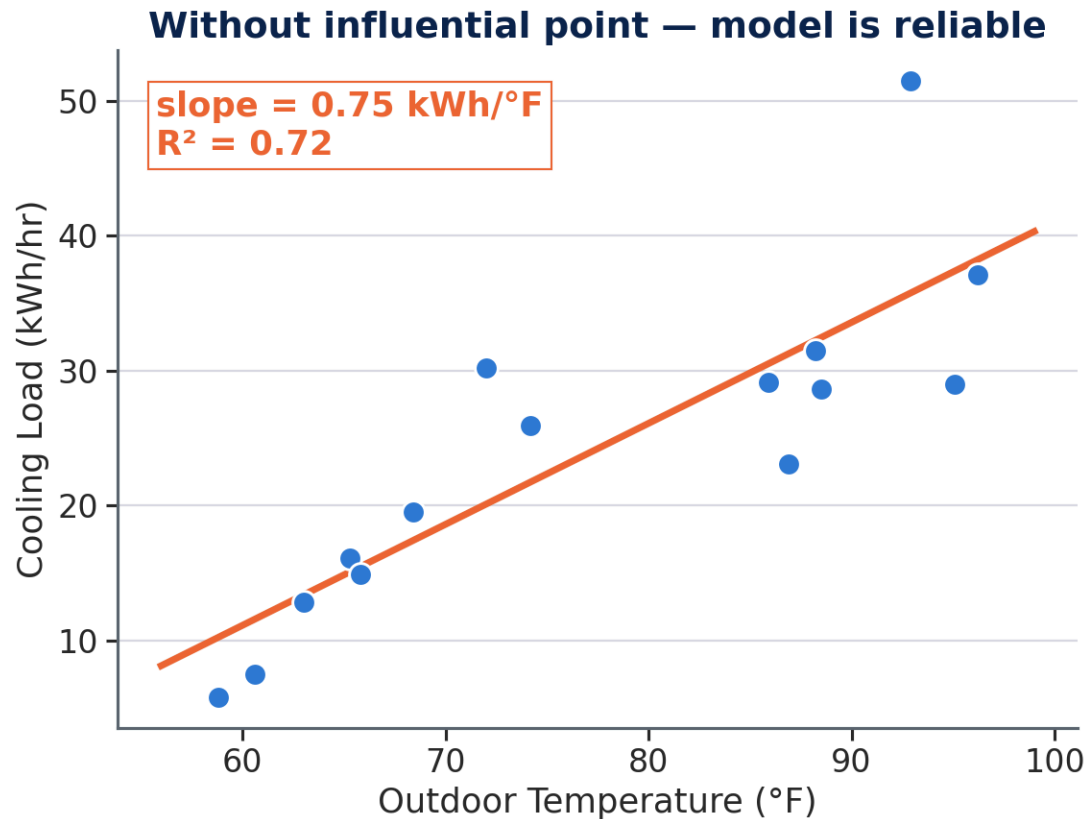
What a good residual plot looks like:

- Residuals scattered randomly above and below zero
- No obvious pattern across the range of x

What signals a problem:

- Systematic positive or negative residuals in a particular x-range → model is biased in that region
- Fan-shaped spread → the model becomes less reliable at high temperatures

Outliers and Influential Points



One observation at (120°F, 0 kWh) cut the slope from **0.78 to 0.25 kWh/°F** — making the model nearly useless.

Outliers – Three Concepts

Term	What it means	Effect on OLS
Outlier	Unusual y value at a typical x	Inflates residual sum; may bias slope
High-leverage point	Extreme x value (far from $\bar{x}$)	Pulls the line toward itself
Influential point	Unusual y AND extreme x	Can completely change the slope

An influential point combines both conditions — the figure above is an example.

Always inspect your data for extreme points before reporting any model. In small datasets, one bad sensor reading or data entry error can invalidate the entire analysis.

Tucson Cooling Model

Putting the numbers together

The Fitted Model

$$\overline{\text{Cooling_kWh}} = -60.68 + 1.1585 \times \text{Outdoor_Temp_F}$$

Model statistics:

- Slope $\beta_1 = \mathbf{1.1585 \text{ kWh/}^\circ\text{F}}$
- Intercept $\beta_0 = \mathbf{-60.68 \text{ kWh}}$
- $R^2 = \mathbf{0.601}$
- p-value ≈ 0 — if temperature had no real effect, a slope this large would be essentially impossible by chance (Week 11 formalizes this)
- $n = 8,760$ hourly observations

Sanity check — plug in the mean:

- Mean outdoor temp: $\bar{x} = 68.7^\circ\text{F}$
- Predicted: $-60.68 + 1.1585 \times 68.7 = \mathbf{19.0 \text{ kWh/hr}}$
- Actual mean cooling: $\mathbf{19.0 \text{ kWh/hr} \checkmark}$

The regression line always passes through $(\bar{x}, \bar{y})$.

Prediction at Design Conditions

Sizing question — ASHRAE 1% design

- Design temperature: **108°F** (top 1% of Tucson hours)
- Prediction:
 - $-60.68 + 1.1585 \times 108 = \mathbf{64.4 \text{ kWh/hr}}$
- Annual mean: 19.0 kWh/hr
- **Peak-to-mean ratio: 3.4×**

Sizing for the mean would leave the building severely under-cooled on hot summer days.

Control question — economizer cutoff

- ASHRAE 90.1 economizer limit: **75°F** (Climate Zone 2B)
- Below 75°F: outside air provides free cooling
- Prediction:
 - $-60.68 + 1.1585 \times 75 = \mathbf{26.2 \text{ kWh/hr}}$
- Above 75°F: mechanical cooling required every hour

Regression Appears Across CE Disciplines

The same model form recurs wherever one continuous quantity drives another.

Discipline	Predictor x	Response y	Slope meaning
HVAC (this week)	Outdoor temp (°F)	Cooling load (kWh/hr)	+1.16 kWh per °F
Structural	Applied load (kN)	Mid-span deflection (mm)	spring stiffness — Hooke's Law
Transportation	Daily traffic (vehicles)	Pavement wear (ESAL/yr)	load damage factor
Water resources	Annual rainfall (in)	Annual runoff (in)	runoff coefficient
Construction	Engineer's estimate (\$M)	Winning bid (\$M)	0.89 — the average bid ratio

The regression you learned this week is not an HVAC tool — it is a universal CE tool. The method travels; only the domain vocabulary changes.

Uncertainty in the Prediction

The slope 1.1585 kWh/°F is an **estimate** — computed from 8,760 observations with noise.

Every estimate has uncertainty. `scipy.stats.linregress` returns the **standard error of the slope**.

One call fits the line and returns the slope, intercept, r , p -value and the **standard error of the slope** — all five at once.

Quantity	Value
Slope β_1	1.1585 kWh/°F
Standard error of β_1	0.0101
95% confidence interval	1.139 to 1.178 kWh/°F

The interval is $\beta_1 \pm 1.96 \times \text{SE}$. It is narrow because $n = 8,760$ — with that much data the slope is pinned down tightly even though R^2 is only 0.601.

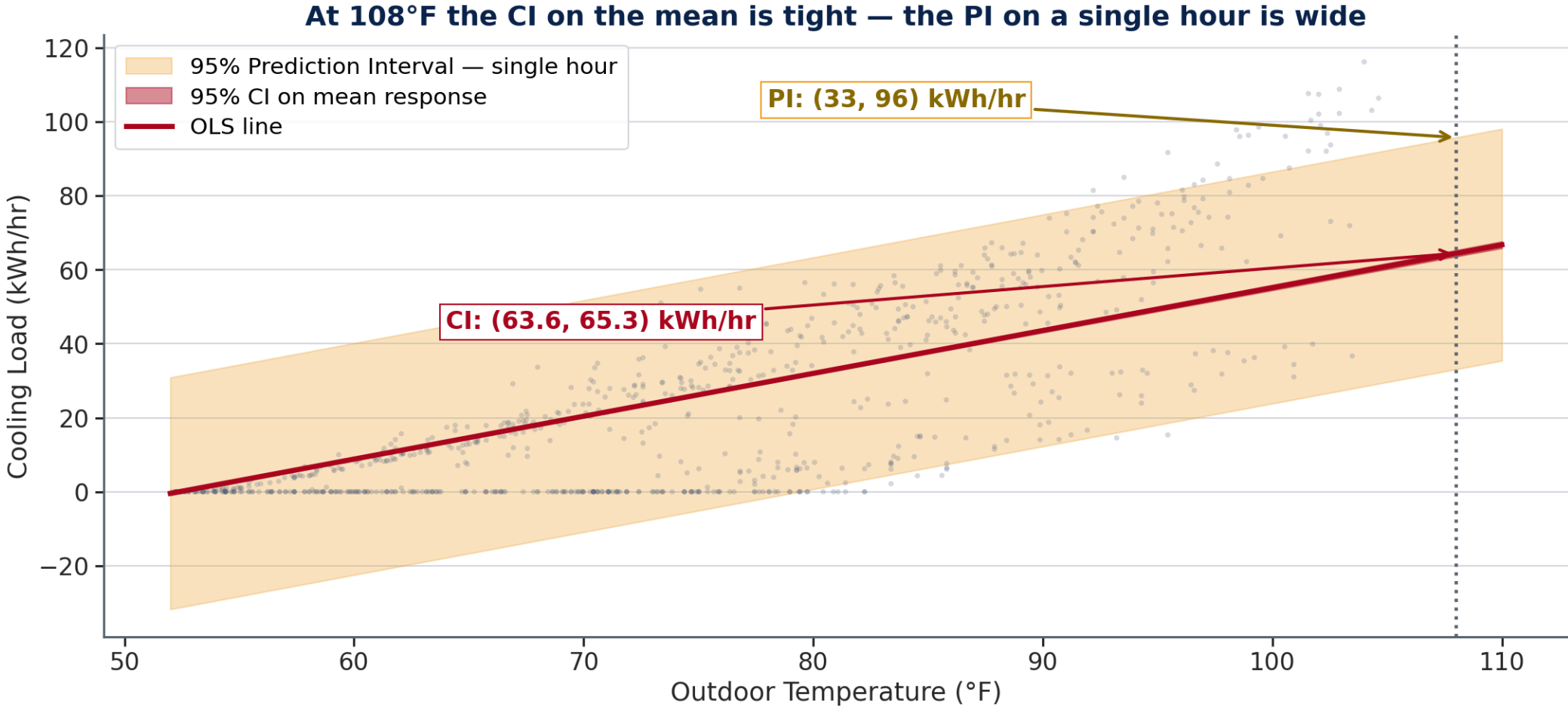
The tutorial covers how to call it and read the output.

Uncertainty in the Prediction — Interpretation

Interpretation: We are 95% confident the true slope lies between **1.139 and 1.178 kWh/°F**.
At 108°F, this 0.039 range translates to only ± 0.8 kWh/hr of slope uncertainty in the final prediction.

💡 The slope CI is narrow because $n = 8,760$ is large. But the uncertainty in a *single predicted hour* is far larger — see the next slide.

Confidence Band vs. Prediction Band



Prediction Interval — The Engineering Deliverable

Two bands exist around a regression line. For equipment sizing, only one matters.

Band	What it bounds	At 108°F
95% CI on mean	Where the average cooling falls for many 108°F hours	(63.6, 65.2) kWh/hr — tight
95% Prediction Interval	Where a single future 108°F hour is likely to fall	(30, 99) kWh/hr — wide

Why so wide? $R^2 = 0.601$ means 40% of variance is unexplained — occupancy, solar gain, plug loads, equipment schedules. Any single hot hour could produce 30 to 99 kWh of cooling demand.

⚠ A chiller sized at 64 kWh/hr matches the point prediction but fails the upper end of the prediction interval. Sizing to the *mean response* means the system is undersized half the time it hits 108°F. The prediction interval upper bound — not the point estimate — is the engineering conservative case.

Design Decision — Which PI Bound?

90 seconds, then discuss

The 95% **Prediction Interval** at 108°F is (33, 96) kWh/hr.

You are sizing a new chiller. The unit costs **\$800 per kWh of capacity**.

Question A: Which bound do you use to size the chiller? Why?

Question B: What is the dollar cost of choosing the upper bound vs. the lower bound?

✓ Use the Upper PI Bound — 96 kWh/hr

A — Size to the upper bound: 96 kWh/hr (conservative design)

Design choice	Capacity	Capital cost	Risk
Lower bound (aggressive)	33 kWh/hr	\$26,400	Near-certain failure at peak
Mean estimate	64 kWh/hr	\$51,200	Fails ~50% of peak hours
Upper bound (conservative)	96 kWh/hr	\$76,800	Low — designed for 97.5th-percentile hour

B — Cost of conservatism: $\$76,800 - \$26,400 = \$50,400$

That \$50,400 is the insurance premium against a heatwave failure in July.

In professional practice: engineers size to the upper bound, then apply an additional safety factor on top.

Where the Model Breaks Down

Physical constraint: cooling energy cannot go below 0 kWh (chiller off = no cooling).

The linear model predicts negative values when:

$$-60.68 + 1.1585 \times T < 0 \implies T < 52.4^\circ\text{F}$$

In the dataset:

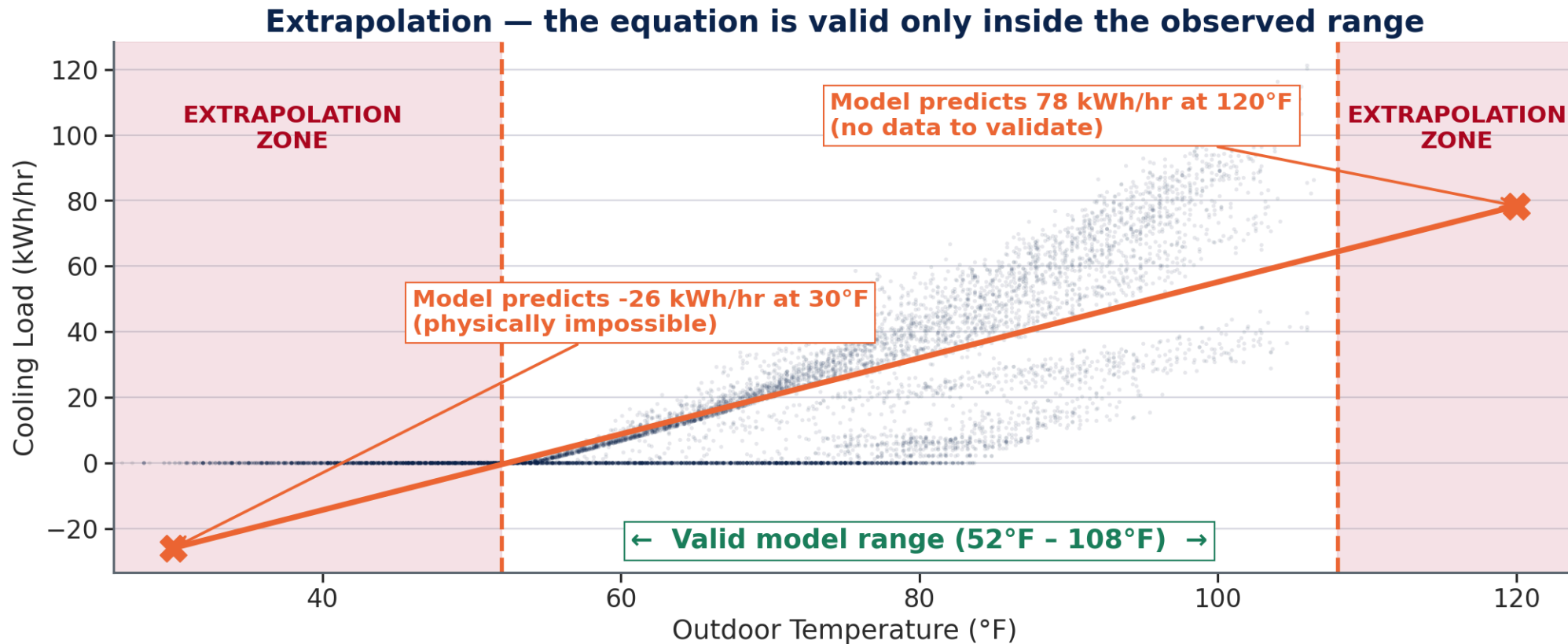
- 3,702 hours with Cooling_kWh = 0 (chiller off, ~42% of year)
- All occur when outdoor temp < ~52°F
- Model predicts negative cooling for those hours

What the residual plot shows:

- At $T < 45^\circ\text{F}$: all residuals are **positive**
- Actual = 0, Model predicts negative $\rightarrow$ residual = 0 - (negative) > 0
- Systematic positive bias at low temperatures

This is a structural failure of the linear model, not noise.

Extrapolation — When the Line Lies



Use the model only within the **observed data range** ($\approx 52^{\circ}\text{F} - 108^{\circ}\text{F}$). Outside that range, the equation has no physical foundation.

Improving the Model — Preview of Week 10

One fix: condition on occupancy before fitting.

Model	R ²	What it captures
All hours (all-year)	0.601	Temperature effect, mixed with chiller-off hours
Business hours only	0.910	Temperature effect with occupancy held constant
Summer only	0.562	Temperature effect, narrower range

During business hours, internal heat sources — lights, computers, people — and occupancy are roughly constant — outdoor temperature explains **91%** of the residual variance.

Week 10: add `BusinessHours` as a second predictor explicitly:

$$\hat{y} = \beta_0 + \beta_1 \cdot \text{Outdoor_Temp_F} + \beta_2 \cdot \text{BusinessHours}$$

From Regression to Machine Learning

The same idea at larger scale

The AI Landscape — Where Regression Fits

Artificial Intelligence — techniques that let computers emulate human behavior: learn, decide, recognize patterns, solve problems.

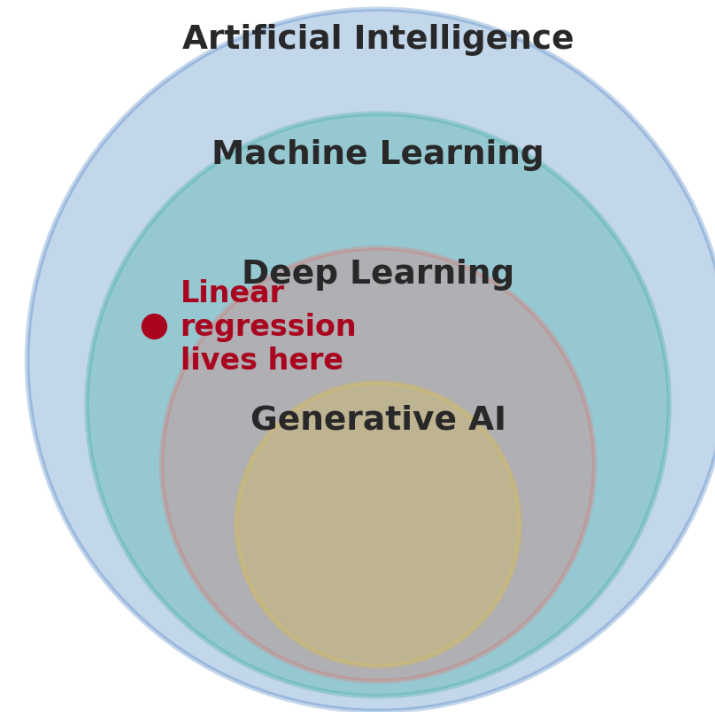
Machine Learning — a subset of AI.

Algorithms that detect patterns in large data sets and adapt, using supervised or unsupervised methods.

Deep Learning — a subset of ML that stacks layers of neural networks to extract high-level features from raw input.

Generative AI — a subset of DL that produces new content — text, images, code — from a prompt.

Each ring is a subset of the one outside it



Regression as Supervised Learning

Linear regression is the simplest **supervised learning** algorithm.

Statistics term	Machine learning term
Response variable y	Label / target
Predictor variable x	Feature
Regression slope β_1 / Intercept β_0	Weight/coefficient / Bias term
Least squares	Loss function minimization
R^2	Model performance metric
Residual plot	Error analysis / diagnostics

Every supervised ML model — from logistic regression to neural networks — generalizes the same idea: **find parameters that minimize prediction error on labeled training data.**

Three Paradigms of Machine Learning

Supervised Learning

Learn from **labeled examples**

($x \rightarrow y$ pairs)

- Regression — predict a number
- Classification — predict a category

Unsupervised Learning

Find **structure without labels**

- Clustering — group similar items
- Dimensionality reduction (PCA)

Reinforcement Learning

Learn by **trial and reward**

- Agent takes actions, receives feedback
- Game playing, robotics

Deep Learning — What Makes It Different

Traditional ML (including OLS regression):

- **You engineer the features** — you decide outdoor temperature is the predictor
- Model finds the best coefficients for those hand-crafted features

Deep Learning:

- **The network learns its own features** from raw data (pixels, tokens, signals)
- Stacked layers of transformations, each learning progressively abstract patterns

Deep Learning — What Makes It Different (cont.)

	Traditional ML	Deep Learning
Features	Hand-crafted by domain expert	Learned automatically from data
Data requirement	Hundreds to thousands of rows	Millions to billions
Interpretability	High (β_1 has physical units)	Low — "black box"
CE use cases	Regression, prediction, classification	Computer vision, NLP, control

Large Language Models — Regression at Massive Scale

An LLM is trained to **predict the next word (token)** — a supervised prediction problem.

The mathematical framework is identical to OLS:

- OLS: find β_0, β_1 to minimize prediction error on (x, y) pairs
- LLM: find billions of weights to minimize prediction error on (text $\rightarrow$ next word) pairs

	OLS Regression	Large Language Model
Parameters	2 (β_0, β_1)	~100 billion
Training data	8,760 observations	Trillions of tokens
Training time	Milliseconds	Weeks on thousands of GPUs
Output	$\hat{y} = \beta_0 + \beta_1 \cdot x$	Probability distribution over next token

Week 9 Quick Reference

Concept	Formula
Regression equation	$\hat{y} = \beta_0 + \beta_1 x$
Residual	$e_i = y_i - \hat{y}_i$
R^2 (from r)	r^2
Slope units	units(y) / units(x)
Predict at x^*	$\beta_1 \times x^* + \beta_0$

Tucson model numbers:

- $\beta_1 = 1.1585 \text{ kWh/}^\circ\text{F}$
- $\beta_0 = -60.68 \text{ kWh}$
- $R^2 = 0.601$
- Prediction at $108^\circ\text{F} = 64.4 \text{ kWh/hr}$
- Prediction at $75^\circ\text{F} = 26.2 \text{ kWh/hr}$

Quick Check — Before the In-Class Exercise

45 seconds for A, 60 seconds for B

Question A: The regression model predicts cooling = $-60.68 + 1.1585 \times 50 = -2.8$ kWh/hr at 50°F. Is this a problem? Why or why not?

Question B: If $R^2 = 0.601$, what percentage of cooling variance is NOT explained by outdoor temperature? Name one physical factor that likely accounts for part of it.

Quick Check — Answers

Before the in-class exercise

Question A: -2.8 kWh/hr at 50°F is a **model limitation, not a calculation error** — cooling energy cannot go below zero (the chiller turns off). The model is valid above $\sim 52^{\circ}\text{F}$; below that it extrapolates into impossible territory.

Question B: If $R^2 = 0.601$, then **39.9% of cooling variance is unexplained** by outdoor temperature alone. Physical factors that account for the residual:

- Occupancy schedule (lights, equipment, people)
- Solar gain and internal plug loads (cloud cover, servers, refrigeration)
- Building thermal mass (lag between outdoor T and indoor response)

 $R^2 = 0.601$ is solid for a single predictor; Week 10 adds occupancy and reaches $R^2 = 0.910$.

Three Deliverables This Week

	What it covers	When
Weekly Quiz	15 questions on this lecture and the Tutorial	Closes 8:00 AM Tuesday, Oct 27
In-Class Exercise	The Colab notebook — all sections, plots, and written answers	Work Wed + Fri · due Wed 9:00 AM next week
Homework	A separate take-home problem set	On your own · due Wed 9:00 AM next week

The exercise spans two sessions. You start it Wednesday after the tutorial, and finish it Friday. **You are not expected to finish in class** — use the session time for the parts where having me in the room actually helps, and complete the rest on your own.

💡 Three submissions, three D2L Assignments folders. The quiz closes 8:00 AM Tuesday, Oct 27; the exercise and the homework are both due Wednesday 9:00 AM — no late work is accepted.

Looking Ahead — Week 10: Multiple Regression

Today: one predictor $\rightarrow \hat{y} = \beta_0 + \beta_1 \cdot \text{Outdoor_Temp_F}$

Week 10: two predictors $\rightarrow \hat{y} = \beta_0 + \beta_1 \cdot \text{Outdoor_Temp_F} + \beta_2 \cdot \text{BusinessHours}$

New questions Week 10 answers:

- After accounting for temperature, does occupancy explain additional variance?
- How much does the cooling baseline shift between occupied and unoccupied hours?
- Can adding more predictors hurt a model? (Adjusted R^2 vs. R^2)

Before next class: in your in-class exercise notebook, predict whether adding `BusinessHours` will increase R^2 . Write one sentence explaining your reasoning.